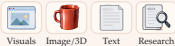


a

Disagreement-based preference mining

Open-ended tasks



Existing datasets New outputs

Metrics and judges

Signal 1 Signal 2 Sample
 $A > B = A > B$ Less

$A > B \neq B > A$ More

Conflict → sample more
 + random sample



Reuse labels
or ask humans
for new ones

Human feedback

Signals can share errors

b

Depth-first evaluator evolution

Reject the candidate, retain the direction and diagnostics



One direction, successive revisions

Example: make layout errors count in the final score

Base model fixed

Candidate

Complete program



Example edit
Add a layout
error detector

No agreement gain

Not accepted

Retain diagnostics



Observed in execution

Layout errors detected.
Final scores unchanged.

Agent's next question

Do these findings reach
the scoring step?

Next revision

Complete program



Example edit
Pass findings
into scoring

Run and inspect again

Keep diagnostics after rollback

Test each candidate

End direction → start another

Candidate acceptance



Same checking set

Better agreement + passed checks
→ update best program

No gain → keep best program



Best program

c

Downstream applications

Output selection



Rank and select

Critic-guided refinement



Reward-guided training



Reward
Planned

Constructed example, not an observed trajectory

Best program fixed after search